

EXPERIMENT NO.2

AIM:- Calibration of energy meter by rotating type substandard meter

INSTRUMENTS REQUIRED :-

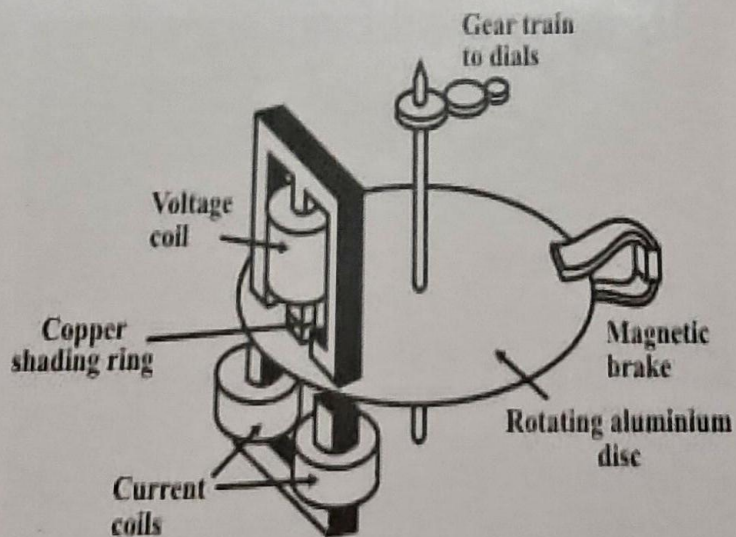
Instrument	Type	Range
Ammeter	MI	0-10A
Voltmeter	MI	0-300V
Energy meter	-	-
Substandard meter	-	-
Connecting leads	-	16Nos.

THEORY:-

The current coils of energy meter under test and substandard test are connected in series with the load Hereafter knowing constants of both meters the error for meter under test can be calculated.

Constant of substandard meter = 1000 rev/kwhr

Constant of energy meter = 600 rev/kwhr



Watt-hour meter.

A single phase induction type energy meter consists of driving system, moving system, braking system and registering system. Each of the systems is briefly explained below.

Driving system: - This system of the energy meter consists of two silicon steel laminated electromagnets. M1 & M2 as shown in fig. ! The electromagnet M1 is called the series magnet and the electromagnet M2 is called the shunt magnet. The series magnet M1 carries a coil consisting of a few turns of thick wire. This coil is called the current coil (CC) and it is connected in series with the circuit. The load current flows through this coil. The shunt magnet M2 carries a coil consisting many turns of thin wire. This coil is called the voltage coil (VC) and is connected across the supply it consist of current proportional to the supply voltage. Short circuited copper bands are provided on the lower part of the central limb of the hunt magnet.

By adjusting the position of these loops the shunt magnet flux can be made to lag behind the apply voltage exactly 90° . These copper bands or called power factor compensator (PFC).

A copper shading band is provided on each outer limb of the shunt magnet (fc1 & fc2) these band provides frictional compensation.

Moving system: - The moving system consists of a thin aluminum disc mounted on a spindle and is placed in the air gap between the series and the shunt magnets. It cuts the flux of both the magnet forces are produced by the fluxes of each of the magnets with the eddy current induced in the disc by the flux of the other magnets. Both these forces act on the disc. These two forces constitute a deflecting torque.

Braking system: - The braking system consists of a permanent magnet called brake magnet. It is placed near the edge of the disc as the disc rotates in the field of brake magnet eddy current are induced in it. These eddies current react with the flux and exert a torque. This torque acts in direction so that it' opposes the motion of disc. The braking torque is proportional to the speed of the disc.

Registering system: - the disc spindle is connected to a counting mechanism this mechanism records a number which is proportional to the number of revolutions of the disc the counter is calibrated to indicate the energy consumed directly in kilo watts-hour (kWh)

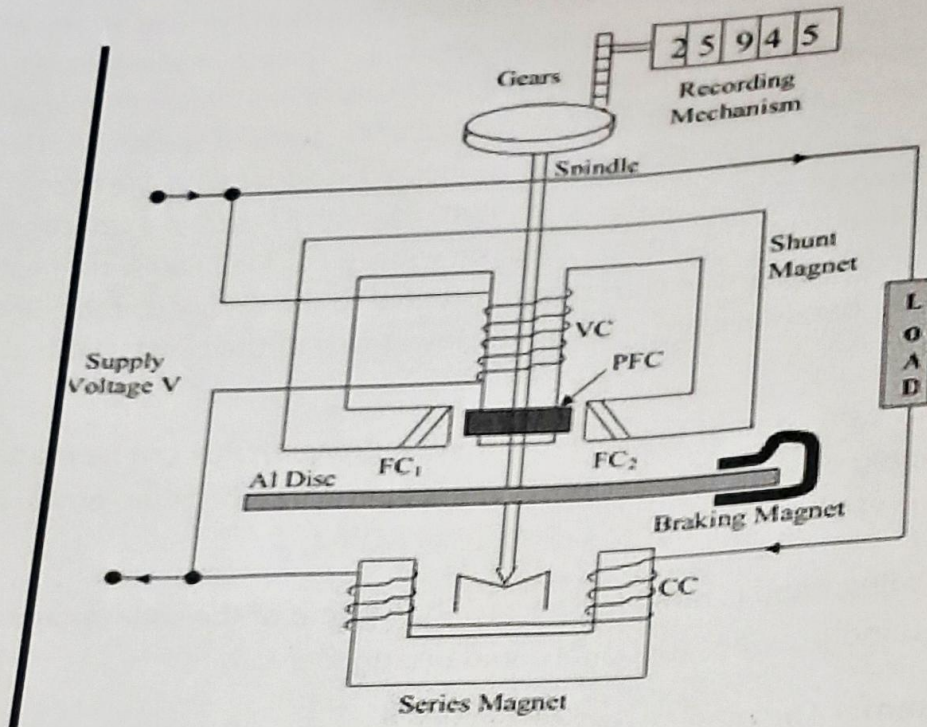


Fig.1

FC1=Fiction Compensators

PFC = Power factor compensator

CC = Current coil

VC=Voltage coil

REGISTERING OR RECORDING SYSTEM

The disk spindle is connected to a counting mechanism. This mechanism records a number which revolution of the disk the counter is calibrated to indicate the energy consumed directly in kilowatt-hour (KWH)

In this experiment the purpose is to calibrate the energy meter. It means we wish to find out the error in the energy meter. This calibration is possible only if some other source or instrument to know the reading available. Here we are calibrating this energy meter with the help of voltmeter, Ammeter and wattmeter.

The actual energy consumed

$$= VIt \text{ watt-Sec.}$$

$$= (VIt / 3600 \times 1000) \text{ KWh}$$

This time 't' is measured in seconds for a given number for revolution (say 2) of the disk of the energy

Indication type energy meters have a constant marked on the meter. It is in terms of number of revolutions per KWh

Let us say 750 revolutions per KWH. Then the energy recorded by the energy meter is given by $13/50$ KWH. The error in the instrument can be calculated as under

$$\text{Error} = \text{Actual energy} - \text{Recorded energy.}$$

A graph of error vs the load current is plotted. This is also known as the calibration curve of the energy meter.

Voltmeter 0-300

Ammeter 0-10A

Variac 230/0-270, 10A

DIAL TEST

Adjust the load resistance so that 8-9 A flows in the circuit to flow half an hour. Take voltmeters and ammeter readings at regular intervals say each 5 minutes. Note the dial reading in the beginning of the test and at the end of half an hour. The difference of the two readings give the energy indicated by energy meter. Calculate the actual amount of energy consumed by using

$$P = (VI \ 0.5 / 1000) \text{ KWhr}$$

Calculate error % age using

$$\% \text{ age} = 100 \times \text{Actual Energy} - \text{Recorded Energy} / \text{Actual Energy}$$

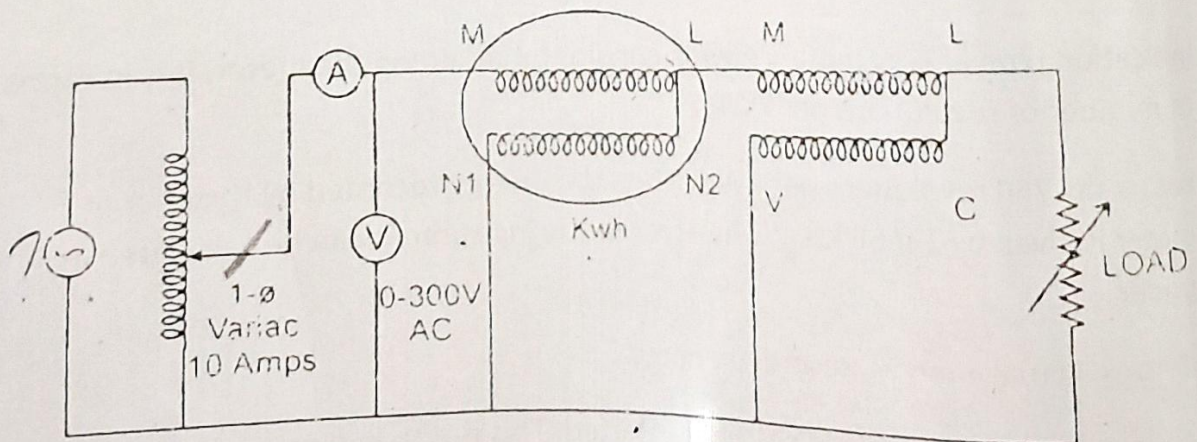


Fig. (2) Circuit diagram for Energy Meter Testing

PROCEDURE

Connect the measuring instruments to the circuit diagram terminal. Connect Voltmeter to the voltmeter terminals, Ammeter to the Ammeter terminals, Wattmeter to the Wattmeter terminals on the panel respectively with help of patch cord.

Connect energy meter to the energy meter terminals on the panel circuit diagram with correct polarity 1 M.L.N1 and N2 to the panel terminals M, L, N1 and N2 respectively with help of patch cord.

Also connect load to the load terminals on right hand side of panel and single phase AC power supply of 230V/50HZ

Keep the load OFF Position and switch ON the power supply and Move the MCB/DP in upward direction i.e 'ON' position & adjust voltage 230V by means of variac.

CALIBRATION OF ENERGY METER :-

Now put some load say 1000W disk of energy meter starts Impulse at A certain speed.

Record the time taken for Impulse with the help of stop watch.

Take Voltmeter and Ammeter readings

Repeat this process for more number of readings and note down these readings in observation table.

DIAL TEST:-

Adjust the load so that a current of flows in the circuit. Allow this current to flow for One hour

Take voltmeter and ammeter readings at regular intervals say after 5 minutes.

Note the dial reading in the beginning of the test and at the end of one hour.

The difference of the two readings give the energy consumed by using indicated by energy meter Calculate the actual value of the energy consumed by using

$$E = (V \times I / 1000) \text{ KWhrs}$$

Calculate the error % using

$$\% \text{age} = 100 \times \text{Actual Energy} / \text{Recorded Energy}$$

OBSERVATION TABLE

V (Voltage)	I (Current)	t(time for impulse)	Actual Energy VI KWhr	Recorded Energy KWh	Error

PRECAUTIONS

1. All meters should be connected in correct polarity.
2. Supply should be switched OFF while making connections.
3. Do not touch terminals on panel while supply is ON.
4. Load should be introduced in steps.
5. Do not exceed beyond rated values.
6. All connections should be tight and clean.
7. The number of revolutions of the disk of the energy meter should be counted with reference to the red mark on the disk.